CAMERA CONTROLLER — signal flow

GND is a global net on every sheet. Double-click a block to open it.

1. Battery 2S + protection



File: sheets/power_in.kicad_sch

2. Buck to 5 V



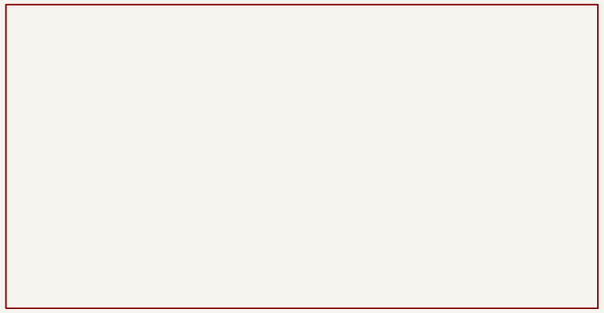
File: sheets/buck_5v.kicad_sch

3. ESP32 + camera



File: sheets/esp32_camera.kicad_sch

4. COTS / SRAD supplies



File: sheets/supplies.kicad_sch

Block 4 is electrically isolated from the rest of this board on purpose.

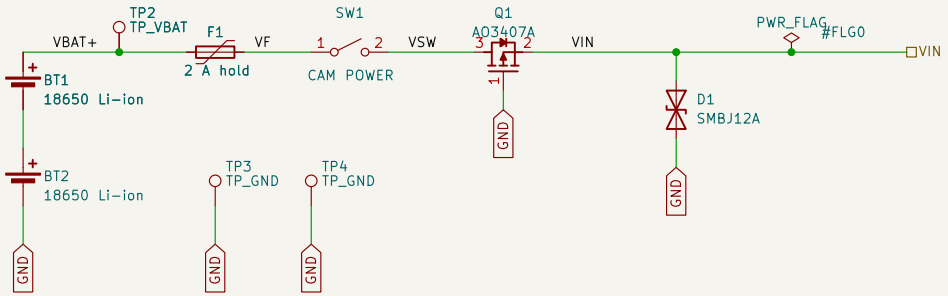
Stanford SSI		
Sheet: /		
File: battery_pcb_30pin.kicad_sch		
Title: Battery PCB — AV bay power, arming and camera		
Size: A3	Date:	Rev: A
KiCad E.D.A. 10.0.0		Id: 1/5

2 x 18650 in series: 7.4 V nominal, 8.4 V full. Charge the cells off the board, one at a time, in a Li-ion charger.

F1: 2 A PTC right at the pack +. Q1: reverse-polarity PMOS (AO3407A, drain to the pack side, gate to GND; about 0.05 V at the 0.6 A load).

D1: TVS, 12 V standoff (above the 8.4 V full pack), clamps below 20 V; everything on VIN is rated 25 V or more.

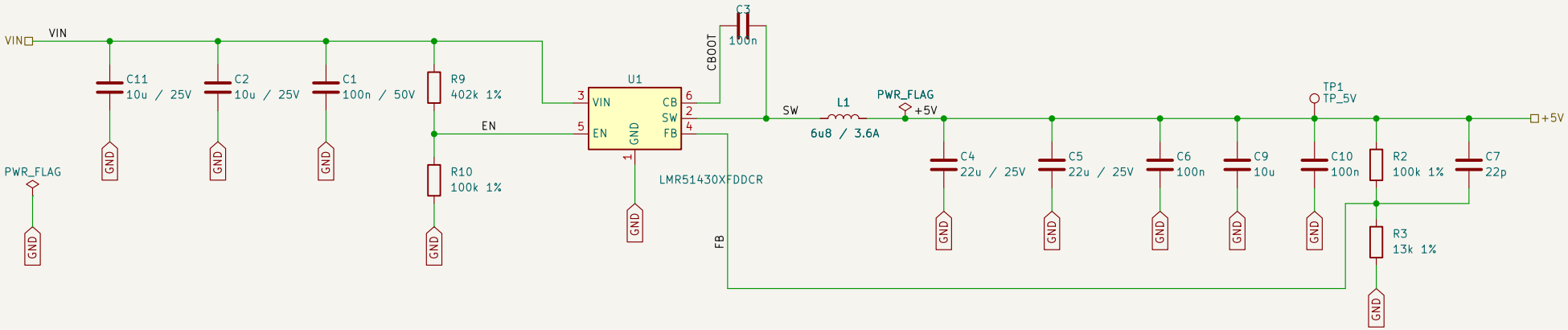
Low-battery cut-off: U1's EN divider on sheet 2 (about 2.7 V per cell).



Stanford SSI		
Sheet: /1. Battery 2S + protection/		
File: power_in.kicad_sch		
Title: 1. Camera + ESP32 supply (2S)		
Size: A4	Date:	Rev: A
KiCad E.D.A. 10.0.0		Id: 2/5

LMR51430 buck, 500 kHz: VOUT = 0.6 x (1 + 100k / 13k) = 5.22 V

Input caps: C1 (100 nF) sits right at U1's VIN/GND pins; C2, C11 (10 uF) are the bulk.
UVLO (R9/R10 on EN): on above 1.23 V x (1 + 402k/100k) = 6.2 V, off below 1.08 V x 5.02 = 5.4 V at VIN,
so the camera pack stops at about 2.7 V per cell instead of being run flat (U1's own UVLO is 3.6 V).
Output caps: C4, C5 (22 uF) and C6 (100 nF) sit at L1 / U1. C9 (10 uF) and C10 (100 nF) sit at the DevKitC's 5V pin (J1 pin 19) on the board.



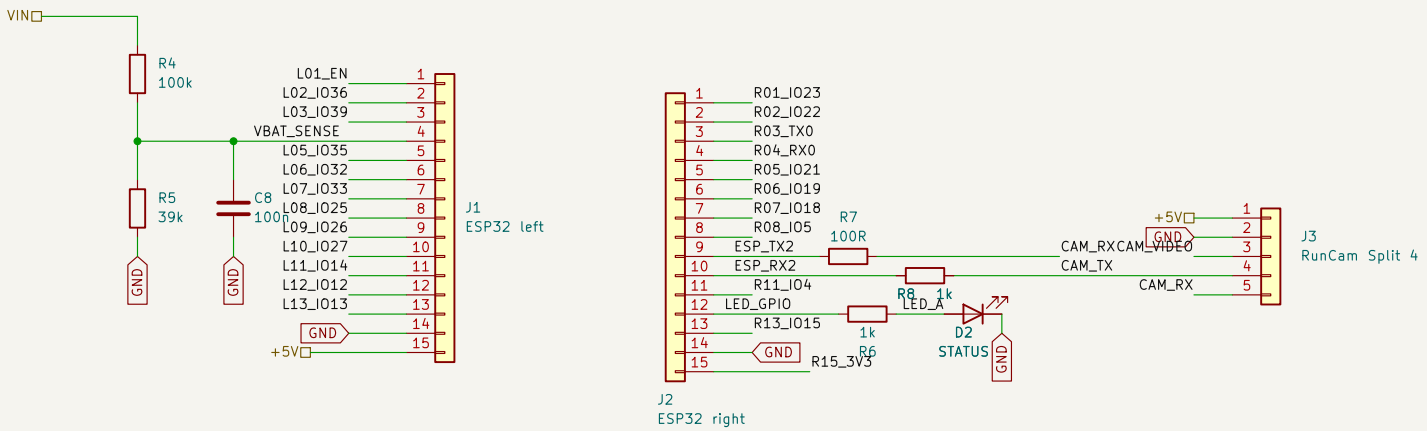
Stanford SSI		
Sheet: /2. Buck to 5 V/		
File: buck_5v.kicad_sch		
Title: 2. Buck 8.4 V -> 5.0 V		
Size: A4	Date:	**Rev: A**
KiCad E.D.A. 10.0.0		Id: 3/5

2 x 1x15 female header for a 30-pin ESP32 DevKit V1 (ESP-WROOM-32, DOIT pinout: EN ... VIN / D23 ... 3V3). Pin 1 is at the antenna end.

Pack sense: VBAT_SENSE = VIN x 39k / (100k + 39k). Full pack 8.4 V (VIN about 8.35 V after Q1) -> 2.34 V at GPIO34, inside the ESP32 ADC's 2.45 V range (11 dB).

Firmware: VIN = VBAT_SENSE x 139 / 39; pack = VIN + about 0.35 V.

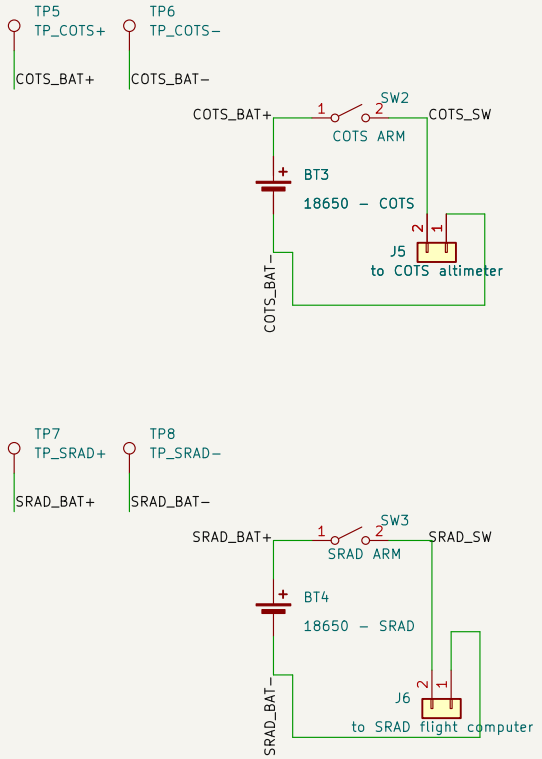
RunCam Split 4: 5-20 V in. Single 5-pin harness: power, ground and UART (R7 / R8 in series with the UART lines).



Stanford SSI		
Sheet: /3. ESP32 + camera/		
File: esp32_camera.kicad_sch		
Title: 3. ESP32 DevKit V1 (30-pin) socket + RunCam Split 4		
Size: A4	Date:	Rev: A
KiCad E.D.A. 10.0.0		Id: 4/5

Own cell, own switch, per DTEG 6.9.1. SW2 / SW3 take the pin-switch leads; they sit next to the pin-switch stack. All connectors latch (Micro-Fit 3.0).

These returns are deliberately NOT tied to GND — that independence is the point.



Stanford SSI		
Sheet: /4. COTS / SRAD supplies/		
File: supplies.kicad_sch		
Title: 4. Independent 1S supplies		
Size: A4	Date:	**Rev: A**
KiCad E.D.A. 10.0.0		Id: 5/5